Vibration-Based Fault Detection and Classification in Unmanned Aerial Vehicles

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Abstract

Reliable detection of propulsion-system faults is essential for the safe and efficient operation of unmanned aerial vehicles (UAVs). This work investigates a vibration-based fault detection approach using inertial measurement data.

The methodology involves signal preprocessing, frequency-domain analysis using the Fast Fourier Transform (FFT), Power Spectral Density (PSD) estimation, and extraction of statistical and spectral features. These features are used to distinguish healthy operation from selected propulsion fault conditions. The performance of the proposed approach is evaluated using appropriate classification metrics.

Related Work I

- According to previous studies, it was found that multi-resolution transform features given to deep neural networks achieve 91% accuracy for UAV fault classification.
- It was also found that when frequency variations were analyzed with wavelet scattering and long short-term memory (LSTM) models, they managed to achieve upto 99% accuracy in propeller damage detection.
- According to another paper, decision tree and support vector machine algorithms are highly efficient for low processing power applications (It was possible to achieve 93.27% accuracy using 14 features and 98.21% accuracy using 64 features).
- Attempts have been made to carry out real-time onboard diagnosis using accelerometer and gyroscope data with PCA and ANN/SVM models and FPGA-based deep learning networks.

Dataset and Fault Conditions I

Simulated Dataset

- Accelerometer datasets were generated via Monte Carlo simulation to assess the validity of our classification algorithm.
- The simulation produced one-second data windows sampled at 8 kHz, operating at a nominal speed of 6000 RPM.
- The simulated fault classes included a healthy baseline, mass imbalance, shaft misalignment, and bearing defects.
- To test noise robustness, the simulation generated 30 data windows per fault class across varying Gaussian sensor noise levels.

Dataset and Fault Conditions II

External Recorded Dataset

- A DJI Mini 2 Combo quadcopter operating in a steady hover mode at a fixed altitude of 1.2 meters was utilized as a basis for an external validation dataset.
- Vibration measurements were captured using an ADXL335 triaxial analog accelerometer mounted at the geometric center of the UAV, where the four rotor arms intersect.
- For the external dataset, data acquisition was managed by a DAQ-6008 USB device.
- The physical fault conditions tested included a healthy state, mass removal on one blade to simulate mechanical wear, and weight unbalance induced by affixing sticky tape to one blade.
- Each flight session lasted 7 minutes (420 seconds), generating approximately 400,000 raw vibration samples per flight case.

Dataset and Fault Conditions III

- The recorded time-domain signals were fed into Excel files containing time, X, Y, and Z-axis acceleration data.
- The sampling rate for this external recorded dataset was 1024 Hz, which approximately yielded 390, 1-second windows per data file.

Methodology: Signal Preprocessing

Raw discrete accelerometer signals are initially mean-centered. A 4th-order Butterworth high-pass filter with a cutoff frequency of $f_c = 10$ Hz is applied zero-phase to all axes.

This step removes the macro-dynamic wander of the rigid body and the stationary 1g gravity bias on the vertical axis prior to spectral decomposition. Finally, the time-domain signal is multiplied by a Hann window to limit the spectral smearing caused by discontinuities at the window boundaries.

Methodology: Mathematical Models

To model how localized mechanical faults propagate to the flight controller's accelerometer, the UAV chassis is approximated as a Linear Time-Invariant (LTI) mass-spring-damper system.

The structural Accelerance transfer function $H_x(\omega)$ is parameterized using the equivalent mass (m_{eq}), structural damping (c_x), and stiffness (k_x) of the chassis:

$$H_x(\omega) = \frac{-\omega^2}{-m_{eq}\omega^2 + jc_x\omega + k_x} \tag{1}$$

Propeller Mass Imbalance (1/2)

Propeller mass imbalance behaves as a point mass $m_{u,i}$ located at a radial distance r_i for a given motor i. As the rotor spins at an angular velocity $\Omega_i = \frac{2\pi \cdot RPM_i}{60}$, it generates a continuous, rotating centrifugal force.

The total physical forcing function on the horizontal axis is the sum of the individual centrifugal unbalance forces from all N motors:

$$F_x(t) = \sum_{i=1}^{N} m_{u,i} r_i \Omega_i^2 \sin(\Omega_i t + \phi_i) \tag{2}$$

Propeller Mass Imbalance (2/2)

The simulated accelerometer reading $a_x(t)$ is the convolution of this continuous force with the drone arm's structural impulse response $h_x(t)$, superimposed with the parameterized Gaussian sensor noise $\eta_x(t)$ (with variance σ^2):

$$a_x(t) = \left(\int_{-\infty}^{\infty} h_x(\tau) F_x(t-\tau) d\tau \right) + \eta_x(t) \tag{3}$$

The proposed model detects propeller blade damage using vibration signals measured along the X and Y axes of an accelerometer. The recorded signals are divided into non-overlapping 10-second decision windows. Each window is processed using the Fast Fourier Transform (FFT), power spectral density (PSD) estimation, spectral feature extraction, and logistic regression classification.

Rotating Shaft Misalignment

In direct-drive brushless motors, angular misalignment manifests as a rotating bent shaft with a permanent angular offset α. The combined 1Ω radial forcing amplitude F_1 is:

$$F_1 = \frac{(I_p - I_t)\sin\alpha\cos\alpha \cdot \Omega^2}{L_b} + T\sin\alpha \tag{4}$$

Anisotropic shaft stiffness (Δk) and axial bearing preload variations during rotation induce a 2Ω axial phenomenological forcing component F_2:

$$F_2 = \Delta k \cdot \alpha \tag{5}$$

For a multi-motor system, the integrated continuous forcing functions are:

$$F_x(t) = \sum_{i=1}^{N} F_{1,i}\sin(\Omega_i t + \phi_i) \tag{6}$$

$$F_z(t) = \sum_{i=1}^{N} F_{2,i}\sin(2\Omega_i t + \phi_{2,i}) \tag{7}$$

Bearing Imperfections

Bearing defects generate high-frequency periodic micro-collisions. The specific kinematic fault frequencies depend on the number of rollers Z, rolling element diameter d, pitch diameter D, and contact angle α:

$$f_i = \frac{Zf_r}{2}\left(1 + \frac{d}{D}\cos\alpha\right) \tag{8}$$

$$f_o = \frac{Zf_r}{2}\left(1 - \frac{d}{D}\cos\alpha\right) \tag{9}$$

$$f_b = \frac{Zf_r}{d}\left(1 - \frac{d^2}{D^2}\cos^2\alpha\right) \tag{10}$$

The peak impact force F_{max} scales linearly with rotor speed and defect severity:

$$F_{max} = (c_k \cdot \Omega \cdot w_d)\sqrt{m_b K_h} \tag{11}$$

$$F_{bearing,x}(t) = (F_{max}\cos\alpha\cos\beta \cdot \Delta t_{contact}) \sum_{n=0}^{\infty} \delta(t - nT_{fault}) \tag{12}$$

PSD and Regression: Signal Processing

The Discrete Fourier Transform (DFT) of a preprocessed and Hann-windowed frame $x_{q,m}[n]$ is:

$$X_{q,m}[k] = \sum_{n=0}^{N-1} x_{q,m}[n] \exp\left(-j\frac{2\pi kn}{N}\right) \tag{13}$$

The one-sided power spectral density (PSD) estimate for a single frame is computed as:

$$P_{q,m}[k] = \frac{c_k |X_{q,m}[k]|^2}{f_s \sum_{n=0}^{N-1} w^2[n]} \tag{14}$$

Class-Weighted Logistic Regression

The linear score for a scaled feature vector $\mathbf{z}_i$ and the resulting logistic damage score s_i:

$$l_i = \boldsymbol{\beta}^T \mathbf{z}_i + b \tag{15}$$

$$s_i = \frac{1}{1 + \exp(-l_i)} \tag{16}$$

The model minimizes the class-weighted logistic loss with an L2 penalty:

$$\mathcal{L}(\boldsymbol{\beta}, b) = \frac{1}{n_{tr}} \sum_{i=1}^{n_{tr}} \alpha_i [\ln(1 + \exp(l_i)) - y_i l_i] + \frac{\lambda}{2} \|\boldsymbol{\beta}\|_2^2 \tag{17}$$

Frequency-Domain Analysis & Feature Extraction

The expected bin Signal-to-Noise Ratio (SNR) for a pure sinusoidal signal submerged in white noise is:

$$SNR_{bin} = \left(\frac{A^2}{2\sigma^2} \right) \frac{N}{2} \tag{18}$$

The fundamental rotation speed (f_0) is estimated natively from the combined radial spectrum using a harmonic-weighted score function:

$$Score_j = P_r(j) + 0.5P_m(2j) + 0.25P_m(3j) \tag{19}$$

Impulsiveness is quantified using excess kurtosis (Fisher's definition):

$$K = \frac{\frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})^4}{\sigma^4} - 3 \tag{20}$$

Fault Classification

The classification algorithm operates directly on the extracted features via a deterministic decision tree without requiring external RPM telemetry.

1. **Bearing Defect:** Checks envelope spectrum for a fundamental peak and integer harmonic with PFR > 4.0 and excess kurtosis > 0.3.
2. **Healthy Baseline:** If radial 1Ω PFR < 3.5, the system defaults to the Healthy Baseline.
3. **Shaft Misalignment vs. Mass Imbalance:** Ratio of the Z-axis 2Ω harmonic against the radial 1Ω fundamental is calculated. If > 1.0, it is Shaft Misalignment. Otherwise, Mass Imbalance.

Monte Carlo simulations showed 1.00 classification accuracy for healthy, imbalance, and misalignment windows up to a noise level of $\sigma = 0.30\ m/s^2$ (SNR 16.3 dB).

Results: Propeller Mass Imbalance (Time Signals)

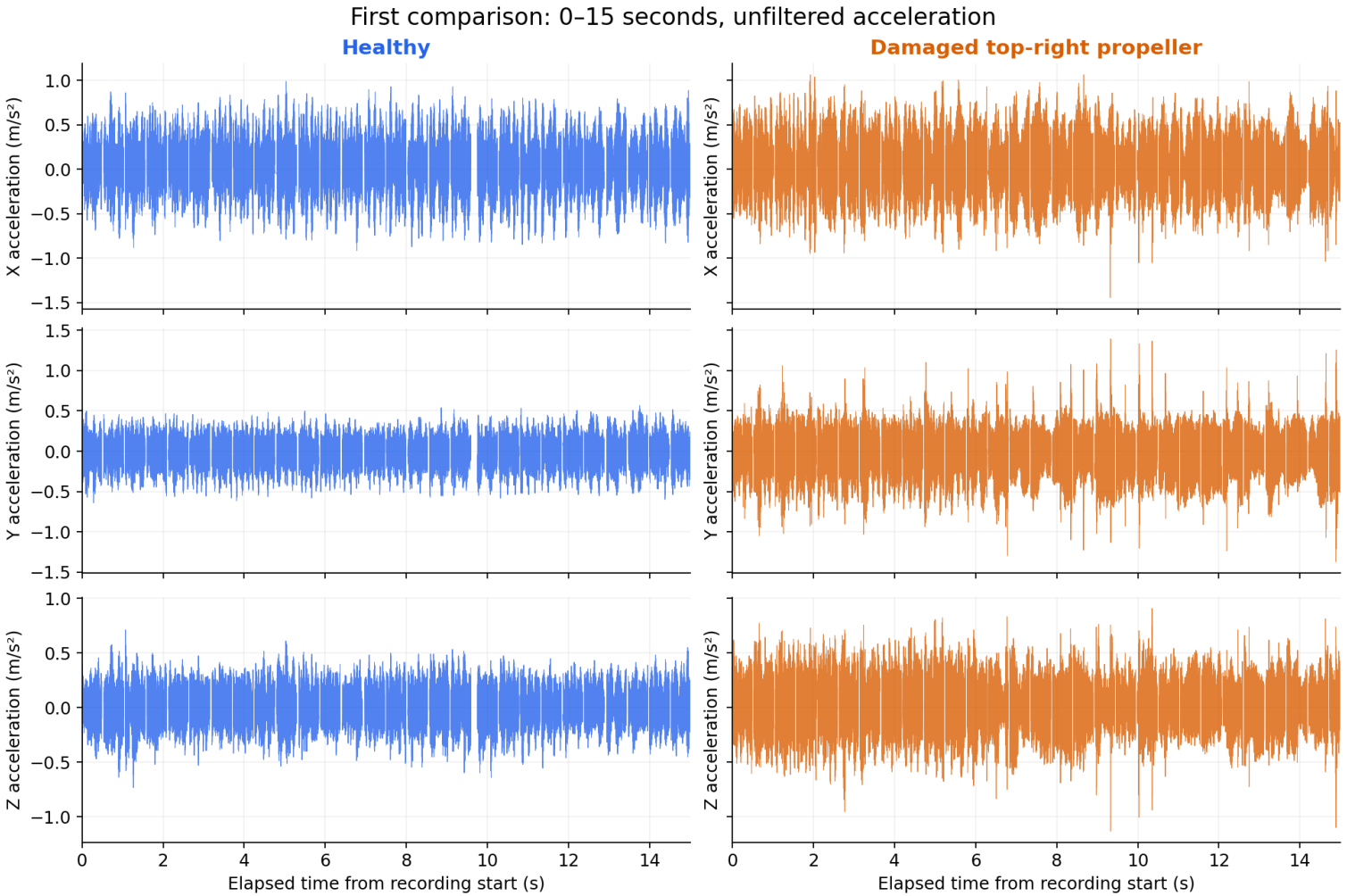


Figure: Comparison of unfiltered triaxial acceleration signals along the X, Y, and Z axes for healthy and damaged-propeller UAV conditions over the first 15 seconds

Results: RMS Comparison

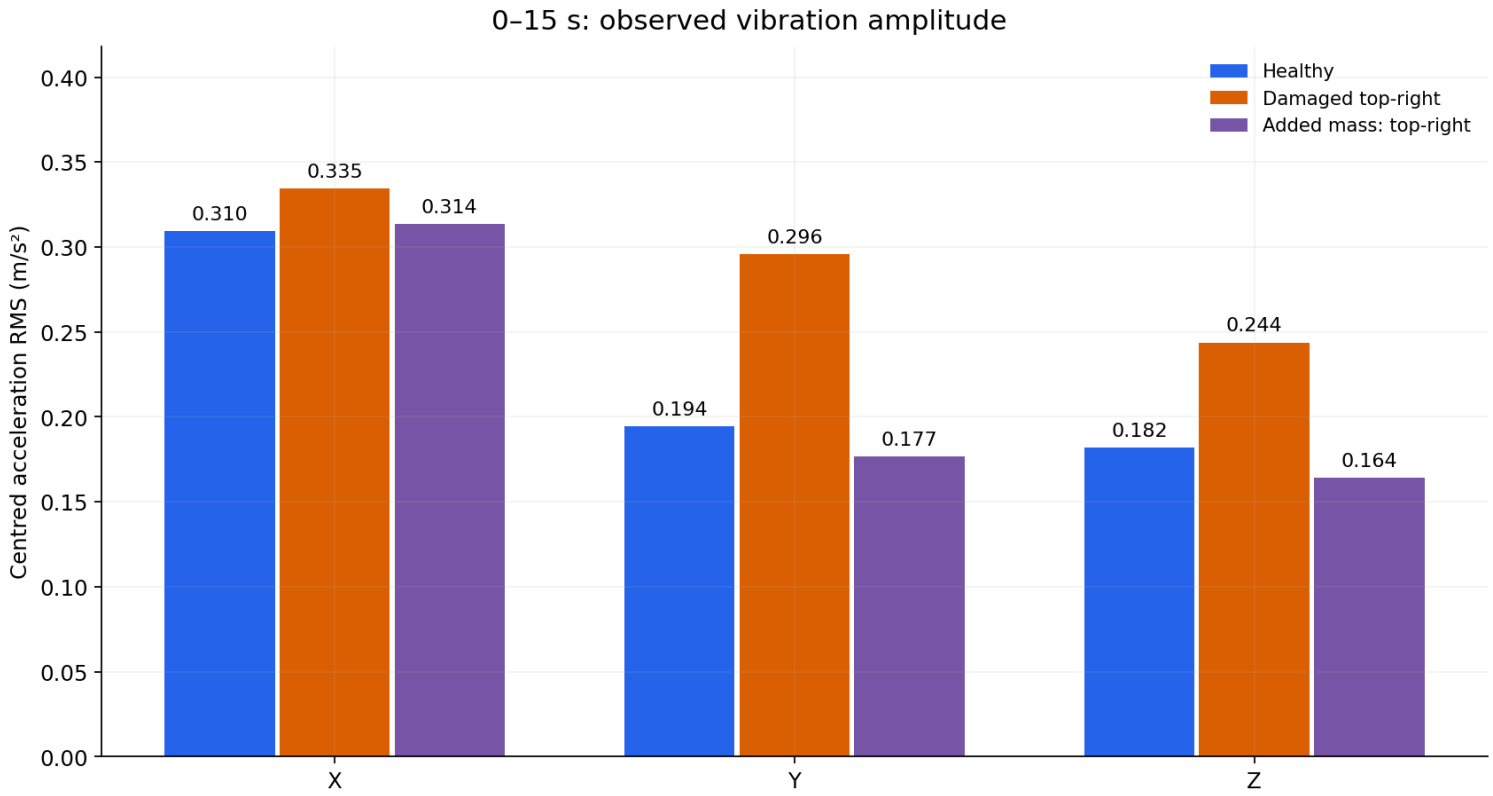


Figure: Comparison of centred RMS acceleration along the X, Y, and Z axes under healthy, damaged-propeller, and added-mass conditions over the first 15 seconds.

Results: Measured vs Synthetic PSD (Damaged)

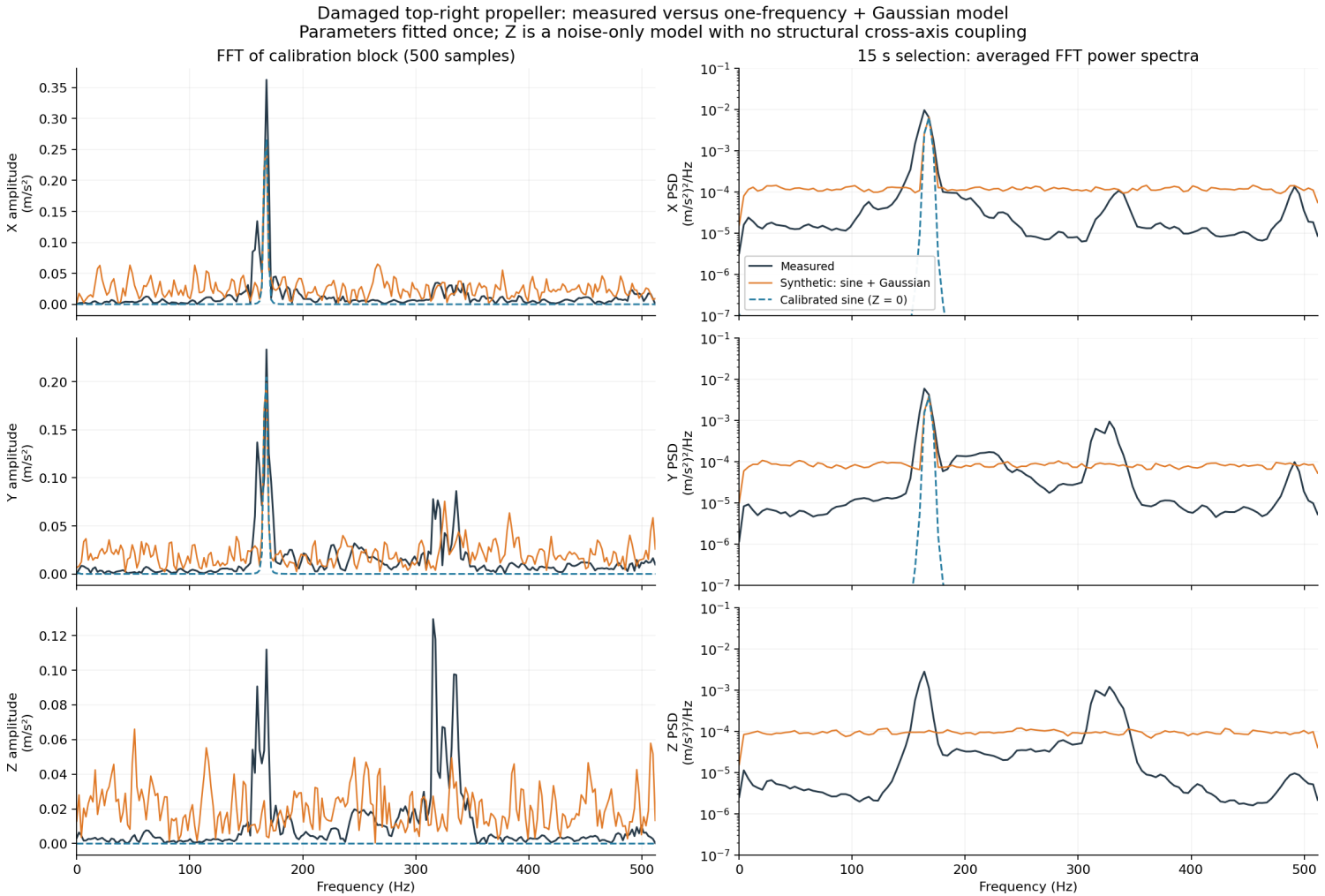


Figure: Comparison of measured and synthetic acceleration spectra for the damaged top-right propeller.

Machine Learning Model Baseline

As observed from the plots, there are differences between damaged and healthy propellers, but visual inspection alone is not robust.

Therefore, a Machine Learning model is developed to detect damage. The model was trained from data of Root Mean Square of PSD of a healthy propeller over a certain time interval, which estimated a baseline for detecting damage. Data of two different drones available on the internet were analysed and two different baselines were obtained for both.

Model Confusion Matrix

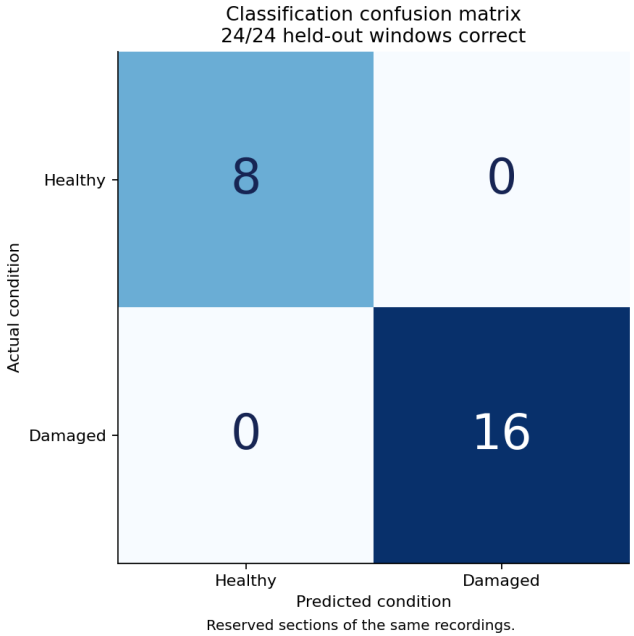


Figure: Confusion Matrix for model of DJI Mini 2 quadcopter

The model correctly classified all 24 held-out windows, including 8 healthy and 16 damaged windows, achieving an accuracy of 100% on this evaluation set.

Model Confusion Matrix

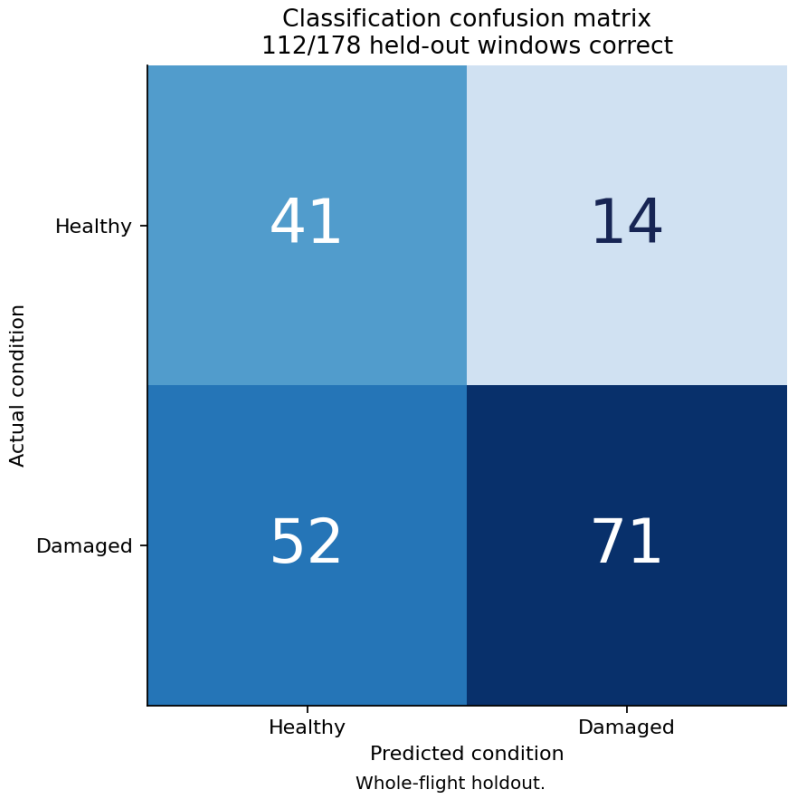


Figure: Confusion Matrix for the second drone

Additional Model Robustness Sweeps

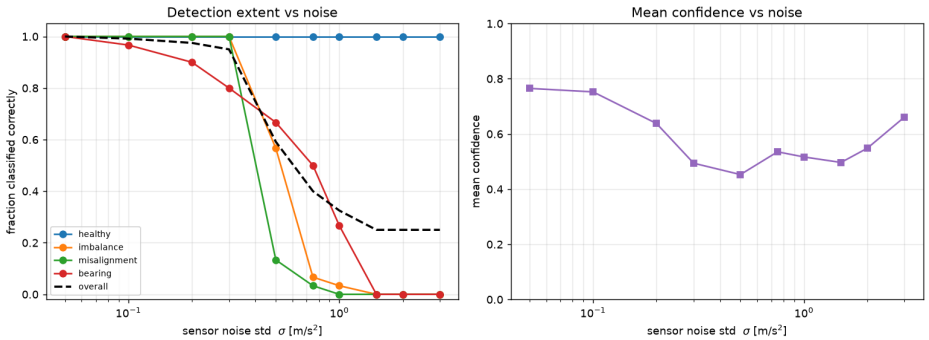


Figure: Noise variation sweeps

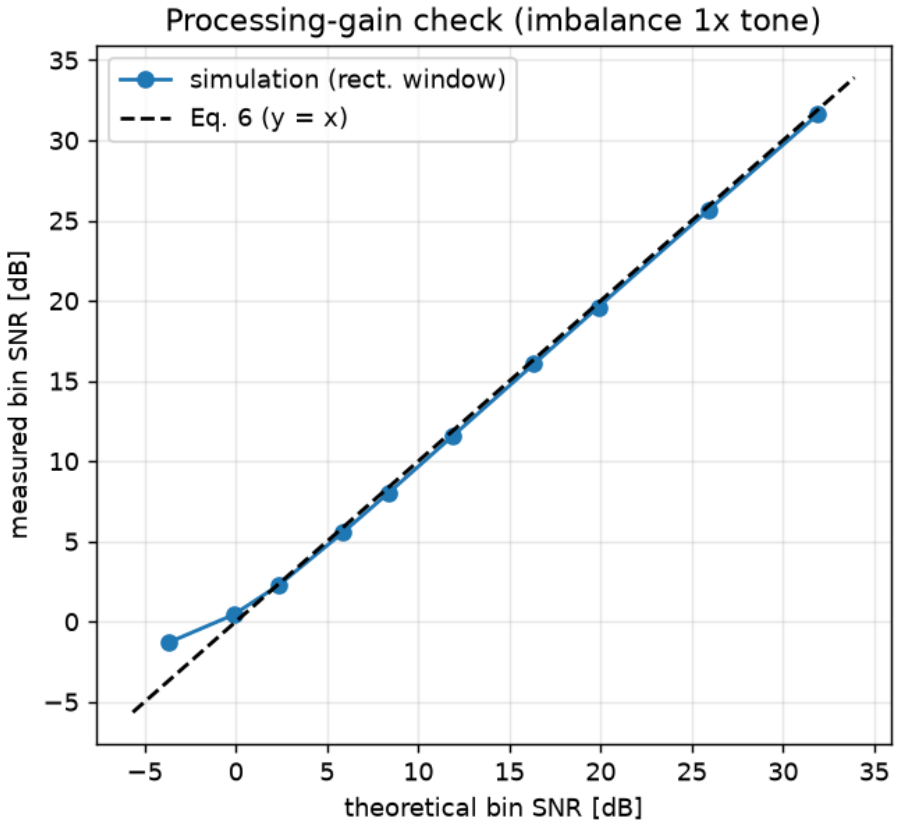


Figure: Theoretical vs Measured SNR

Additional Spectra Analysis

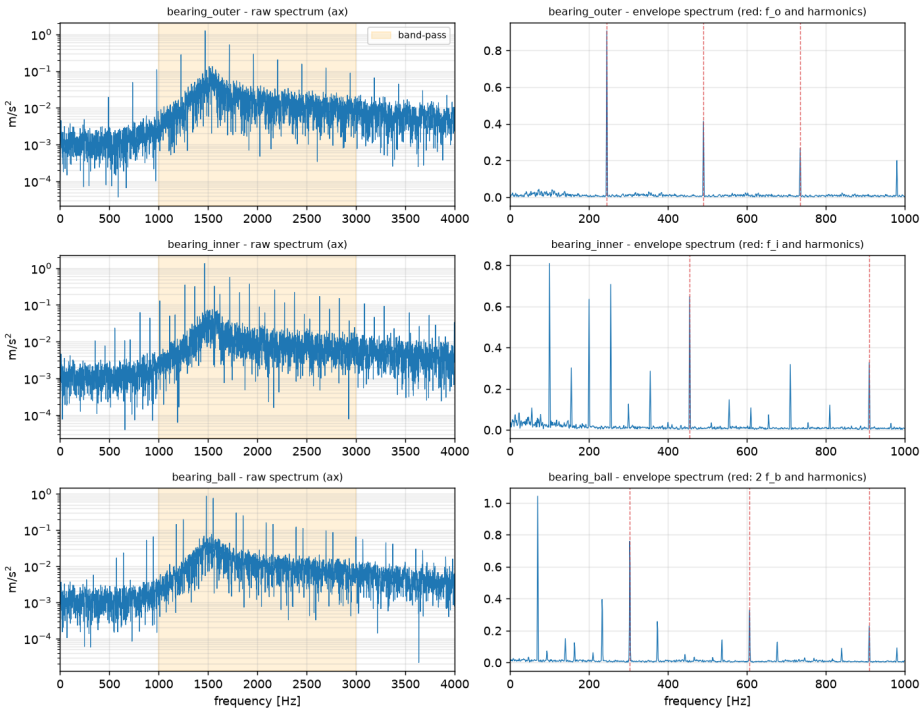


Figure: Bearing defect spectra

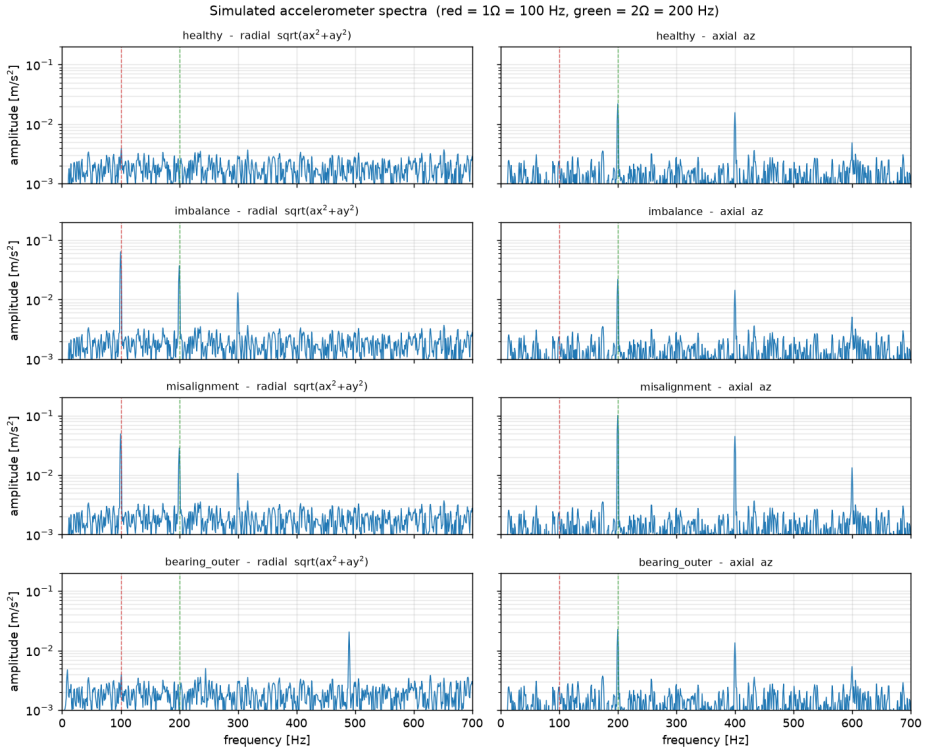


Figure: Mass imbalance spectra

Defect Identification vs RPM

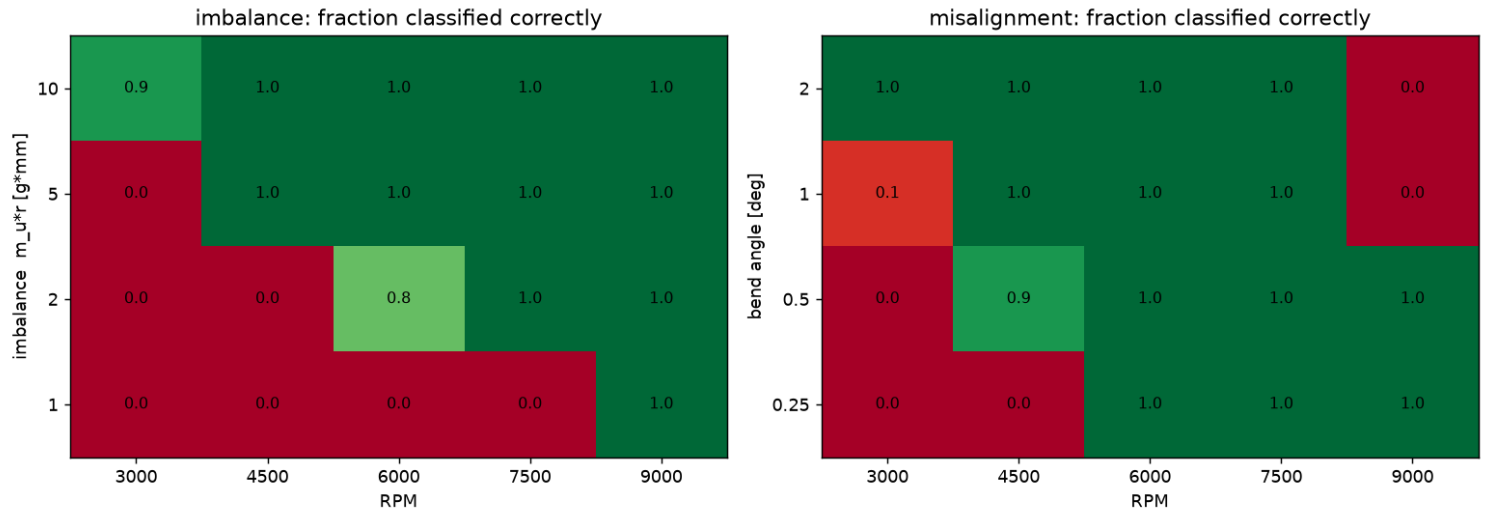


Figure: Variation of fraction of defects identified correctly with operating RPM

Conclusion and Future Work

Our developed equations and fault detection and classification algorithms have been stress tested with various accelerometer data set available online with satisfactory confidence scores. Though there lies some degree of uncertainty and specifically that uncertainty scales with increasing noise, we still have managed to demonstrate the usefulness of signal conditioning and pre-processing on raw data to achieve our results.